

Virtually Being Old : Proteus Effects of Age-Related Avatars on Ageism and Naturalistic Scene Memory in Young Adults

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Figure 1: Examples of personalised avatars depicting the same young (1;3) and older (2;4) individuals.

ABSTRACT

The Proteus Effect (PE) describes behavioural changes driven by stereotypes associated with an embodied avatar. Although age-related embodiment in immersive VR may influence social biases, its effects on episodic memory (EM) remain unexplored. We tested 72 young adults who embodied a personalized young or older avatar, navigated a virtual city with naturalistic scenes, and then completed an incidental scene-memory test. Ageism was assessed before and after the VR session. EM performance was lower in the older-avatar condition, whereas ageism measures showed no reliable change. These findings suggest age-related PE can modulate EM and motivate further work in older adults.

Index Terms: Proteus Effect, Ageism, Stereotypes, Episodic Memory, Virtual Reality, Embodiment.

1 INTRODUCTION

Virtual Reality (VR) represents a promising tool for investigating the impact of stereotypes on human behaviour through embodied avatars, a phenomenon known as the Proteus Effect (PE) [7]. The literature highlights that the PE can be driven by a wide range of avatar characteristics (e.g. age, skin colour, weight, physical ability), influencing behaviour [4], and its potential to mitigate stereotypes [6]. Furthermore, several studies have shown that embodying well-known older individuals, such as an Einstein avatar, can improve performance in planning tasks and reduce ageism (i.e. beliefs, often pejorative, about ageing and older people) compared to embodying a young avatar [1]. As the population ages, the negative impact of ageism on health and well-being of older adults becomes a critical issue. Indeed, a longitudinal study suggest that more negative ageist stereotypes are associated with greater cognitive and physical decline with age, especially poorer episodic memory (EM) [3], defined as the memory of personally experienced events in their spatiotemporal context.

However, the influence of the PE on EM in the context of ageism has never been studied, despite the fact that VR offers high ecolog-

ical validity for the study of EM. Moreover, avatar personalisation is known to enhance embodiment and identification with the avatar [2]. Therefore, this present study aims to investigate the impact of the PE on EM and ageism when embodying a personalised young or older avatar.

2 MATERIAL AND METHODS

2.1 Participants

Seventy-four French-speaking young adults were recruited. Two were excluded for meeting the exclusion criteria, resulting in a final sample size of 72 participants (76 percent of women, mean age = 21.3 ± 3.9 , mean years of education = 13.4 ± 1.8). They were equally assigned to the young ($N = 37$) and older ($N = 35$) personalised avatar conditions. Groups did not differ on baseline neuropsychological measures, including episodic memory, cognitive flexibility, self-concept, and trait anxiety.

2.2 VR equipment, virtual environment, and avatars

Participants were equipped with a head-mounted display (HTC Vive Pro) and real-time full-body tracking (hands, feet, hips) during the VR session. The virtual environments were developed using Unity 2019.2.5f1. The avatars were created thanks to Character Creator 4, which enabled facial personalisation based on an ID-style photograph of each participant. Body shape and clothing were identical across conditions, but the older avatar's face and body were visually aged (wrinkles, liver spots and grey hair).

2.3 General procedure

During the VR session, participants were assigned to one of two avatar conditions (young or older ; Fig. 1). They were first placed in front of a mirror to embody their personalised avatar and familiarize with it during a **visuo-motor task** lasting approximately two minutes. Then, they were all exposed to two counterbalanced **stereotype-inducing rooms** aiming to increase the salience of the ageist stereotypes. In the first room, participants explored an old-fashioned bedroom (old wooden furniture, 1920s' song, etc.) and observed themselves in a mirror. In the second room, they explored a sports room while wearing actual weights (handles and ankles) and then practiced a sports exercise. After that, participants underwent an **incidental scene-encoding memory task** by navigating in a virtual city. They had to follow a predefined route, with instructions only to take photographs and explore the environment as a visitor. Along the route, they encountered 15 naturalistic daily-life

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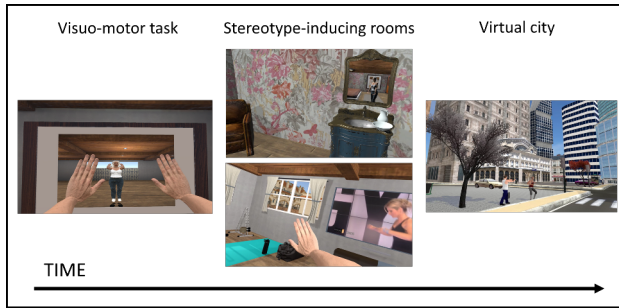


Figure 2: Time course of VR tasks.

scenes featuring objects, animals or characters, each accompanied by visual and auditory animations (e.g. two men dancing hip-hop to American rap music) (Fig. 2).

Measures of implicit (Implicit Association Task; IAT) and explicit (Fraboni Scale of Ageism; FSA) ageist stereotypes were collected before and after the VR session, allowing pre-post change scores to be computed. Embodiment (Peck and Gonzales-Franco Embodiment Questionnaire), sense of presence (Igroup Presence Questionnaire) and cybersickness (Simulator Sickness Questionnaire) were assessed after the virtual rooms and the virtual city. Participants then completed a standardised verbal free recall test [5], in which they asked to recall all scenes they could remember (**what**) in as much **details** as possible (e.g. colours, number of characters, verbatim elements), and to specify **where** each scene occurred in space (in front, at left, at right) and **when** it occurred in time (beginning, middle, end). This procedure yielded scores for memory content (what, details, where and when) and a **binding** score (i.e. interconnection between what, details, where and when) for each scene. Finally, they completed a recognition task by indicating which scenes they recognized (what). One week later, participants completed the FSA again and underwent a delayed verbal free recall test identical to the immediate test.

3 RESULTS

The main analyses consisted of a series of between-subjects ANOVAs comparing conditions (older vs. young avatar) on all memory outcomes and stereotype change scores. Embodiment, sense of presence, and cybersickness did not differ across conditions and were therefore not included as covariates in the analyses.

3.1 Scene memory

Immediate free recall. The results show a significant effect of condition on the number of scenes recalled (**what** ; $F(1,69) = 5.17$, $p = .03$, $n^2 = 0.07$; Fig. 3), the spatial location (**where** ; $F(1,69) = 5.76$, $p = .02$, $n^2 = 0.08$) and the **binding** ($F(1,69) = 5.17$, $p = .03$, $n^2 = 0.07$; Fig. 3), with better performance after embodiment in the young avatar than in the older avatar, but not for the number of details or the time location (when).

Delayed free recall. The results show a significant effect of condition on the number of scenes recalled (**what** ; $F(1,69) = 4.05$, $p = .05$, $n^2 = 0.06$), with more recalls in the young avatar condition compared to the older avatar condition, but not for the other scores (where, when, details, binding).

Recognition. The results do not show significant effect of condition on the number of events recognised, maybe due to a ceiling effect ($M = 13.62$, $SD = 1.57$).

3.2 Stereotype

The results do not show significance, although IAT change score seems to indicate more stereotype reduction for older avatars ($M = -0.22$, $SD = 0.30$) than young avatars ($M = -0.15$, $SD = 0.28$).

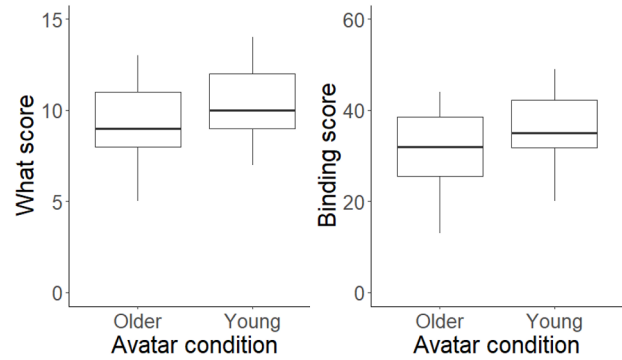


Figure 3: Number of events recalled (what ; at left) and what/when/where/details interconnexion (binding ; at right) by avatar condition (young or older).

4 CONCLUSION

The results of this study show better EM performance when young adults embody their young avatar rather than their older avatar including on the richer memory index (binding), revealing an age-related PE on EM. However, this PE appears to be mainly immediate and transient, as it was observed on binding in immediate free recall but was no longer present at delayed free recall, where it persisted only for factual scene memory (the “what” component). Moreover, the fact that the immediate recognition task was relatively successful in both avatar conditions, may indicate that the PE influences access to memory rather than encoding per se. Future works should consider a longer delay for this task or more scenes to increase the difficulty (old/new) and verify this hypothesis. Otherwise, our initial findings suggest that explicit and implicit stereotypes are not influenced by age-related PE, although a trend seems to emerge in the IAT.

Altogether, this work opens promising avenues to investigate how age-related PE can modulate cognitive processes and ageism, with potential implications for both basic research and targeted interventions.

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